Communication User Authentication and Encrypted Data Transaction Solution

Authentication

Target：

- Robot authenticates request source, only allows authenticated Mobile App to connect.
- Mobile App authenticates request source, only allows authenticated robot to connect.

Encryption Method: the most importation concern of security is the secured storage of key. In this case, symmetric encryption should be adapted and with **AES algorithm**.

A same 128bit AES is stored both in Mobile App and Robot App.

Authentication of request resource in robot

After connected, client side will send request to robot

1. Communication

   Based on “Request-Respond” model

request path: /ubtrobot/communication/auth

param : (String) —— VersionCode, EncryAlgorithm, UserId, salt, Base64.encode(AES(md5(UserId + salt))) //if the user is not login and UserId is empty, set UserId as “ ”, use "," to divide in the String

/// salt can directly use current time stamp ,unit: ms

2. Bluetooth

   Send Request

   Declare type as auth, type

   Data: (String) **VersionCode,EncryAlgorithm,UserId, salt,AES(md5(UserId + salt))**

   Authentication of robot in Mobile App

   Robot authenticates request whether legal or not, if not, reply “failure” and disconnect after 1s; if it is legal, reply “success” with this content:

   param： **VersionCode, EncryAlgorithm, RobotId，salt, AES(md5(RobotId + salt))** , salt can directly use current time stamp ,unit: ms

   Mobile App authenticates robot and disconnect if it is illegal.

Encrypted Communication

Bluetooth communication already encrypted on protocol level and no need for further encryption, this encryption is for Mobile App communication

Data communication structure:

len(4byte) + encryption method(1byte) + encrypted payload

1. Mobile App uses embedded AES key to encrypt the payload
2. Robot App receives encryption method and decrypts the message, also decrypts payload with robot embedded AES key.